DMPK Case Study

In-vitro ADME

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Case Study 1: CYP 2C9 Inhibition (1/2)

The Problem

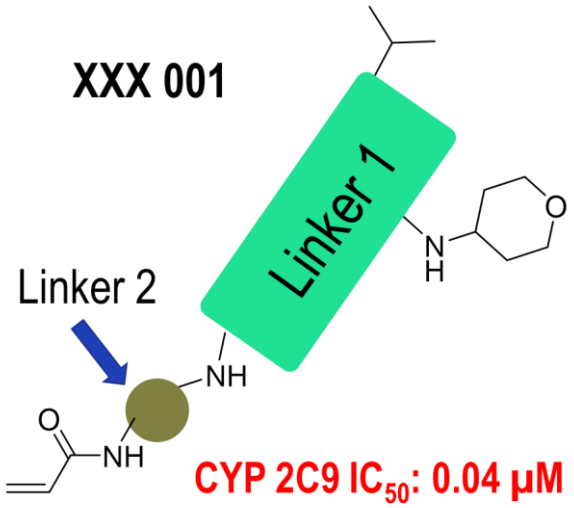
- Active compounds in a project were found to be highly potent inhibitors of CYP 2C9
- The compounds selectively inhibited CYP 2C9 with IC_{50} values <100 nM
- There was no considerable inhibition the other CYP isoforms

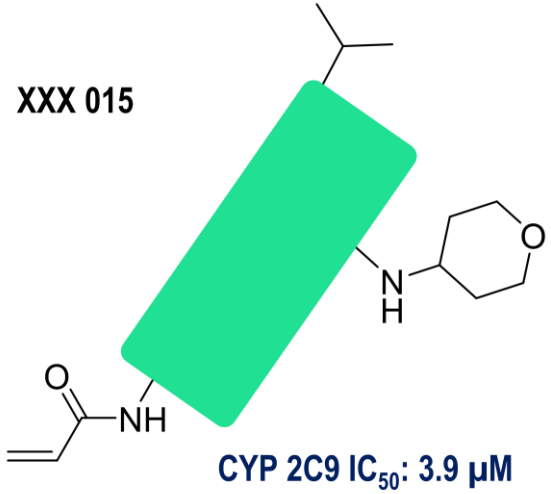
Mitigation Approach

- CYP 2C9 inhibition data was generated for a larger set of compounds with diverse moieties irrespective of their biochemical potency towards the target
- SAR was derived with respect to the CYP 2C9 inhibition

Observations

- Compounds without 'Linker 2' were found to have lower CYP 2C9 inhibition potential
- However, 'Linker 2' was critical for primary activity towards the target





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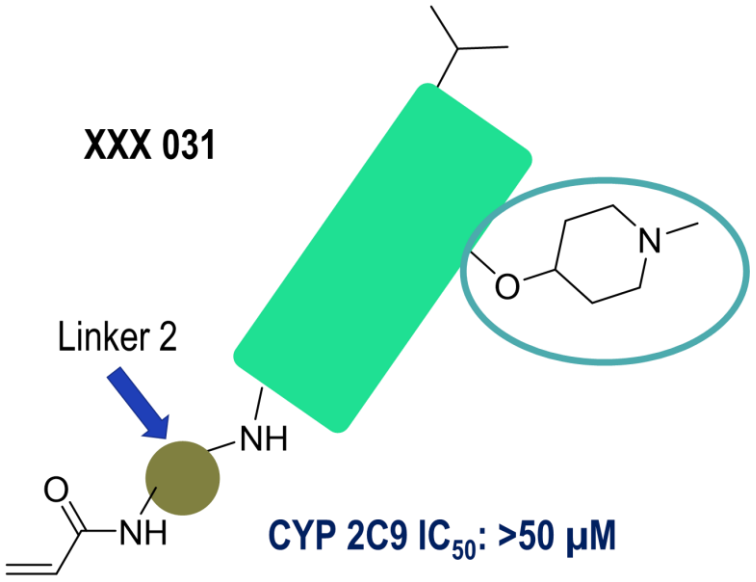
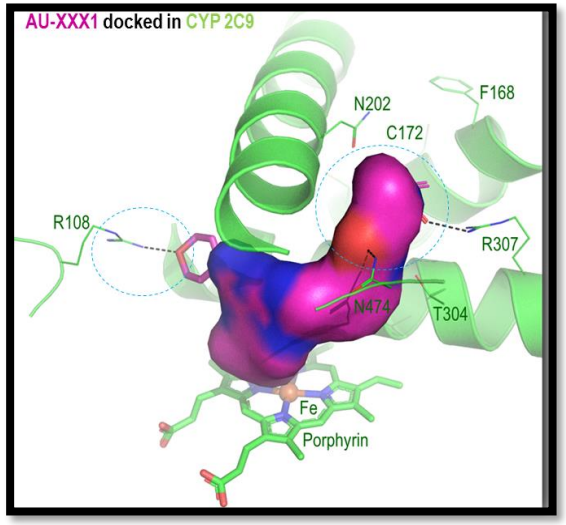
Case Study 1: CYP 2C9 Inhibition (2/2)

Understanding the interaction of the inhibitor with CYP 2C9 was important to mitigate this problem

- CYP 2C9 inhibitors are reported to be involved in a key binding interaction with an Arginine (Arg 108), a cationic residue in the active site of the protein
- This binding site is unique to CYP 2C9 and not present in the other CYP isoforms
- It would require an anionic moiety on the molecule to make this interaction with Arg 108 in CYP 2C9
- Systematic SAR was built on this hypothesis by modifying the anionic groups in different regions of the molecule, meanwhile retaining the primary activity



Replacement of the tetrahydropyron with N-methyl piperidine yielded high CYP 2C9 selectivity, while maintaining the potency towards the target



in-silico docking studies which followed support this finding/hypothesis

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Case Study 02: Plasma Stability of PEGylated Prodrugs:

Background

- The PEGylated Prodrug was designed for delivery to target organs, where it would be converted to Active Drug
- Prodrug conversion in plasma is undesirable

The Problem

- In rat plasma, the prodrug was quickly converted into Active drug and PEG –COOH (PEG acid).
- It was necessary to identify the cause and halt this conversion

Approach

- Customized the plasma stability protocol
- Three types of samples (aqueous, extracted, and post-extracted) were examined instead of single incubation set of samples
- Detected concentrations of drug and byproduct formation by using a short linearity of prodrug, drug and PEG-COOH to authenticate the samples
- Several factors were evaluated to understand the mechanism of the rapid conversion of prodrug to drug in the incubationmixture
- Effect of Temperature, 100% Plasma Vs 10% plasma, different concentrations of prodrug, conducted Mass balance, Effect of Esterase inhibitors like phenylmethylsulphonylfluoride (PMSF) & Sodium fluoride on prodrug conversion was explored

Resolution

- A combination of PMSF and Na_2F esterase inhibitors was found to prevent prodrug-to-drug conversion in rat plasma
- This customized protocol was used to screen the molecules, develop an SAR to understand and overcome the metabolic liabilities.

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